

ASSIGNMENT – 3

1. Create the following users:

☐ alice

☐ bob

☐ charlie

```
ubuntu@ip-172-31-31-42:~$ sudo useradd alice
ubuntu@ip-172-31-31-42:~$ sudo useradd bob
ubuntu@ip-172-31-31-42:~$ sudo useradd charlie
ubuntu@ip-172-31-31-42:~$ sudo groupadd devteam
```

2. Create a group called:

☐ devteam

3. Add users alice and bob to the devteam group.

```
ubuntu@ip-172-31-31-42:~$ sudo usermod -aG devteam alice
ubuntu@ip-172-31-31-42:~$ sudo usermod -aG devteam bob
ubuntu@ip-172-31-31-42:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
```

```
ubuntu@ip-172-31-31-42:~$ cat /etc/group | tail -5
group2:x:1003:
alice:x:1004:
bob:x:1005:
charlie:x:1006:
devteam:x:1007:alice,bob
```

```
ubuntu@ip-172-31-31-42:~$ getent group devteam
devteam:x:1007:alice,bob
ubuntu@ip-172-31-31-42:~$
```

4. Set passwords for all users.

```

ubuntu@ip-172-31-31-42:~$ passwd alice
passwd: You may not view or modify password information for alice.
ubuntu@ip-172-31-31-42:~$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-31-42:~$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-31-42:~$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully

```

```

ubuntu@ip-172-31-31-42:~$ sudo cat /etc/shadow |tail -5
ubuntu!!:20465:0:99999:7:::
deepika:$y$j9T$byQe7sRmVwhkXtvmrI0520$LCxmuAs/o79LyRZbCWKJDKEYuvs0WVxZDKbSTAF2eOD:20466:0:99999:7:::
alice:$y$j9T$10Kt.KXAQacClqhUNqXlc0$GpPSsCGG8B1RptSZ8a.qjAB769C6ZcQDRT935hCJ0iB:20471:0:99999:7:::
bob:$y$j9T$d5kSrte2sBX5Qg8Z0mWxC/$VcxV5cx58er.1Fz9R0ROH2NNFEOE30WwBx3UwVqJmq7:20471:0:99999:7:::
charlie:$y$j9T$.z3BzkaoWxrGg6Tv7qtx51$foLmbDG6hcPUPeqffXeckCIdyUQKeulhfdqGdGz4BvC:20471:0:99999:7:::
ubuntu@ip-172-31-31-42:~$ █

```

5. Verify: ○ User IDs and group memberships using id and groups.

```

ubuntu@ip-172-31-31-42:~$ id alice
uid=1002(alice) gid=1004(alice) groups=1004(alice)
ubuntu@ip-172-31-31-42:~$ id bob
uid=1003(bob) gid=1005(bob) groups=1005(bob)
ubuntu@ip-172-31-31-42:~$ id charlie
uid=1004(charlie) gid=1006(charlie) groups=1006(charlie)

```

```

ubuntu@ip-172-31-31-42:~$ cat /etc/passwd |tail -5
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
deepika:x:1001:1001:Deepika,,,:/home/deepika:/bin/bash
alice:x:1002:1004:./home/alice:/bin/sh
bob:x:1003:1005:./home/bob:/bin/sh
charlie:x:1004:1006:./home/charlie:/bin/sh

```

Part 2: File Ownership & Permissions

1. Create a directory: /opt/projectX

2. Change ownership:

○ Owner: alice

○ Group: devteam

3. Apply permissions such that:

○ Owner & group → full access

○ Others → no access

4. Verify permissions using: ls -l

```

ubuntu@ip-172-31-31-42:/opt$ sudo chown alice:devteam projectX
ubuntu@ip-172-31-31-42:/opt$ ls -l
total 4
drwxr-xr-x 2 alice devteam 4096 Jan 18 13:30 projectX
ubuntu@ip-172-31-31-42:/opt$ sudo chmod 770 projectX
ubuntu@ip-172-31-31-42:/opt$ ls -l
total 4
drwxrwx--- 2 alice devteam 4096 Jan 18 13:30 projectX
ubuntu@ip-172-31-31-42:/opt$ █

```

Part 3: File Attributes (Security Hardening)

1. Create a file inside the directory: config.txt
2. Make the file immutable so it cannot be deleted or modified accidentally.
3. Verify attributes using: lsattr
4. Attempt to delete or edit the file (observe behavior).

```

ubuntu@ip-172-31-31-42:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-31-42:~$ mkdir files
ubuntu@ip-172-31-31-42:~$ cd files
ubuntu@ip-172-31-31-42:~/files$ touch config.txt
ubuntu@ip-172-31-31-42:~/files$ ls
config.txt

```

```

ubuntu@ip-172-31-31-42:~/files$ sudo chattr +i config.txt
ubuntu@ip-172-31-31-42:~/files$ lsattr config.txt
----i-----e----- config.txt
ubuntu@ip-172-31-31-42:~/files$ echo "Hello"> config.txt
-bash: config.txt: Operation not permitted
ubuntu@ip-172-31-31-42:~/files$ rm -r config.txt
rm: cannot remove 'config.txt': Operation not permitted

```

Conceptual Questions

- Why use groups instead of giving permissions to individual users?

Many individuals that work under a same project are put together in a group, so giving access to a group makes it simpler, scalable, more efficient and easier to access the shared resources

- When would an immutable file be useful in production?

An immutable file cannot be modified, deleted, renamed, or overwritten even by root, unless the immutable bit is removed. So this helps in protecting configured files, preventing any tampering against attacks.

- Why is `/etc/shadow` readable only by root?

`Etc/shadow` consists of information of users and their hashed passwords. If others could read it, they could attempt offline password-cracking attacks. Restricting access to root ensures system security and prevents privilege escalation.